Backpressure Fairness in Stream Processing: A Reproducible Native-vs-ROS2 Study

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Abstract

This paper reports a frozen-analysis benchmark of backpressure policy behavior under fairness-sensitive stream stress. Native and ROS2 baseline systems each execute 800 runs (4 policies × 4 workloads × 50 repeats) under a shared run schema. In native results, the rank-1 policy differs by workload: `weighted_drop` (balanced, recovery_stress), `drop_oldest` (minority_spike), and `drop_new` (skewed_burst). Across 72 Holm-corrected pairwise tests among non-error policies, 10 are significant and all significant findings are on `harm_weighted_cost` or `p95_latency_ms`. The `error` policy is retained in reporting and shows zero successful runs in all workloads. ROS2 results are treated as appendix evidence only; under this transport baseline, native means are lower for harm and p95 latency in every comparable workload-policy cell. All numeric claims are traceable to generated JSON/CSV artifacts.

1 Introduction

Queue overflow policy is often selected as an operational tuning parameter. This study evaluates it as a measurable decision variable under fixed stress workloads and fixed analysis rules.

The objective is limited and explicit: compare four overflow policies on latency, reliability, and fairness-sensitive metrics, then report only what is supported by the frozen statistical protocol. The core claim is based on native runtime results. ROS2 transport results are reported in the appendix as supporting comparative evidence under the same run schema.

The paper contribution is an artifact-first report: (1) frozen analysis specification, (2) complete matrix execution records, (3) table-ready outputs linked to run-level JSON/CSV artifacts.

2 Related Work

This manuscript is an empirical artifact report rather than a broad literature survey. The work is positioned at the intersection of stream-performance benchmarking and fairness-sensitive evaluation: the matrix is small and controlled, the metrics are pre-registered, and the interpretation is constrained to observed differences under the frozen protocol.

3 Methods

3.1 Experiment Matrix

We evaluate four policies (`drop_new`, `drop_oldest`, `error`, `weighted_drop`) across four workloads (`balanced`, `skewed_burst`, `recovery_stress`, `minority_spike`). Each policy-workload pair is repeated 50 times with deterministic seed derivation from `seed_base=12000`. The expected matrix size is $4 \times 4 \times 50 = 800$ runs per system.

Table 1: Core performance and reliability metrics.

Workload	Policy	N	Throughput	P95 Latency	Drop	Miss	Harm Cost
balanced	drop_new	50	558.0042	3.4508	0.3333	0.3333	33.7644
balanced	drop_oldest	50	555.0780	3.4743	0.3333	0.3333	33.7675
balanced	error	0	-	-	-	-	-
balanced	weighted_drop	50	558.5661	3.4448	0.3333	0.3333	33.7637
minority_spike	drop_new	50	430.0380	14.1666	0.4000	0.4000	41.3019
minority_spike	drop_oldest	50	430.8375	14.1619	0.4000	0.4000	41.3017
minority_spike	error	0	-	-	-	-	-
minority_spike	weighted_drop	50	425.6626	14.2404	0.4000	0.4000	41.3087
recovery_stress	drop_new	50	289.9619	18.3828	0.5000	0.5000	51.1527
recovery_stress	drop_oldest	50	290.3658	18.3916	0.5000	0.5000	51.1537
recovery_stress	error	0	-	-	-	-	-
recovery_stress	weighted_drop	50	290.4629	18.3720	0.5000	0.5000	51.1518
skewed_burst	drop_new	50	414.6281	14.6361	0.5000	0.5000	51.2351
skewed_burst	drop_oldest	50	410.9764	14.6828	0.5000	0.5000	51.2379
skewed_burst	error	0	-	-	-	-	-
skewed_burst	weighted_drop	50	411.6848	14.6612	0.5000	0.5000	51.2375

3.2 Data Source Freeze

All manuscript numbers are sourced from one frozen artifact root: `outputs/experiments/backpressure_fairn`
No post-hoc experiment rerun or manual number entry is used in this draft.

3.3 Frozen Analysis Specification

Before full execution, `analysis_spec_v1.yaml` fixes (1) metric keys, (2) ranking rule, (3) statistical tests, (4) effect-size thresholds, (5) interpretation constraints.

Primary ranking uses `harm_weighted_cost_mean`; ties are resolved by `event_miss_rate_mean`, then `p95_latency_ms_mean`, then policy id. Pairwise evaluation uses the non-error policy set for six metrics: `harm_weighted_cost`, `event_miss_rate`, `p95_latency_ms`, `group_miss_gap`, `group_latency_gap_ms`, and `drop_rate`.

3.4 Statistics

The protocol uses bootstrap 95% confidence intervals, two-sided Mann–Whitney U tests, Holm–Bonferroni correction ($\alpha = 0.05$), and Cliff's delta. Effect-size interpretation follows fixed thresholds: negligible < 0.147, small < 0.33, medium < 0.474, large ≥ 0.474. Superiority claims are prohibited when Holm-adjusted p is missing or > 0.05.

3.5 Execution Environments

Native stages run in conda environment `schnitzel-stream`. ROS2 stages run with system interpreter `/usr/bin/python3` after sourcing `/opt/ros/humble/setup.bash`, due `rclpy` compatibility constraints. Both systems emit the same run schema contract.

4 Results

4.1 Core Metrics

Table 1 reports 16 workload-policy cells. For non-error policies, each cell has $N = 50$ successful runs. For `error`, all runs fail in every workload (0/50 successful per cell; 200 failed runs total),

Table 2: Fairness-sensitive metrics by policy and workload.

Workload	Policy	Worst Group	Worst Miss	Worst P95	Miss Gap	Latency Gap
balanced	drop_new	group_b	0.3333	3.4630	0.0000	0.0295
balanced	drop_oldest	group_a	0.3333	3.4875	0.0000	0.0289
balanced	error		-	-	-	-
balanced	weighted_drop	group_b	0.3333	3.4586	0.0000	0.0311
minority_spike	drop_new	group_b	0.4000	14.7112	0.0000	8.4356
minority_spike	drop_oldest	group_b	0.4000	14.7097	0.0000	8.4379
minority_spike	error		-	-	-	-
minority_spike	weighted_drop	group_b	0.4000	14.7770	0.0000	8.4396
recovery_stress	drop_new	group_b	0.5000	18.7217	0.0000	7.2337
recovery_stress	drop_oldest	group_b	0.5000	18.7318	0.0000	7.2233
recovery_stress	error		-	-	-	-
recovery_stress	weighted_drop	group_b	0.5000	18.6972	0.0000	7.2034
skewed_burst	drop_new	group_b	0.5000	15.1613	0.0000	8.9927
skewed_burst	drop_oldest	group_b	0.5000	15.1910	0.0000	8.9987
skewed_burst	error		-	-	-	-
skewed_burst	weighted_drop	group_b	0.5000	15.1926	0.0000	9.0154

and these failures are kept in the final tables rather than filtered.

Across non-error policies, mean `drop_rate` and `event_miss_rate` are workload-constant: 0.3333 (balanced), 0.4000 (minority_spike), and 0.5000 (recovery_stress, skewed_burst). Across all non-error cells, `throughput_mean` spans 289.9619 to 558.5661 and `p95_latency_ms_mean` spans 3.4448 to 18.3916.

4.2 Fairness Metrics

Table 2 shows `group_miss_gap_mean`=0.0000 in all non-error cells. The observed `group_latency_gap_ms_mean` ranges from 0.0289 (balanced, drop_oldest) to 9.0154 (skewed_burst, weighted_drop). Pairwise tests for `group_latency_gap_ms` are non-significant in every workload (minimum Holm-adjusted $p = 0.2047$, recovery_stress drop_new vs weighted_drop).

4.3 Policy Ranking

Ranking (Table 3) is workload-specific under the frozen rule: `weighted_drop` is rank 1 for balanced and recovery_stress, `drop_oldest` is rank 1 for minority_spike, and `drop_new` is rank 1 for skewed_burst.

Rank-1 `harm_weighted_cost_mean` with bootstrap 95% CI: balanced `weighted_drop` 33.7637 [33.7631, 33.76 minority_spike `drop_oldest` 41.3017 [41.2998, 41.3037], recovery_stress `weighted_drop` 51.1518 [51.1503, 51.15 skewed_burst `drop_new` 51.2351 [51.2335, 51.2366].

4.4 Graph View

Figure 1 matches the ranking trends in Table 3: the lowest mean harm appears at `weighted_drop` (balanced, recovery_stress), `drop_oldest` (minority_spike), and `drop_new` (skewed_burst).

4.5 Pairwise Significance and Effect Size

Table 3: Policy ranking by workload using fixed tie-breaking rule.

Workload	Rank	Policy	Harm Cost	Miss Rate	P95 Latency
balanced	1	weighted_drop	33.7637	0.3333	3.4448
balanced	2	drop_new	33.7644	0.3333	3.4508
balanced	3	drop_oldest	33.7675	0.3333	3.4743
balanced	4	error	-	-	-
minority_spike	1	drop_oldest	41.3017	0.4000	14.1619
minority_spike	2	drop_new	41.3019	0.4000	14.1666
minority_spike	3	weighted_drop	41.3087	0.4000	14.2404
minority_spike	4	error	-	-	-
recovery_stress	1	weighted_drop	51.1518	0.5000	18.3720
recovery_stress	2	drop_new	51.1527	0.5000	18.3828
recovery_stress	3	drop_oldest	51.1537	0.5000	18.3916
recovery_stress	4	error	-	-	-
skewed_burst	1	drop_new	51.2351	0.5000	14.6361
skewed_burst	2	weighted_drop	51.2375	0.5000	14.6612
skewed_burst	3	drop_oldest	51.2379	0.5000	14.6828
skewed_burst	4	error	-	-	-

Table 4: Pairwise policy significance and effect-size summary (all 72 rows). Wkld: BAL/MNS/RCS/SKB. Metrics: DR/EMR/P95/GMG/GLG/HARM. Policy: DN=drop_new, DO=drop_oldest, WD=weighted_drop, ER=error.

Wkld	Met	Pair	d	p MWU	p Holm	Sig	Best
BAL	DR	DN vs DO	0.0000	-	-	no	DO
BAL	DR	DN vs WD	0.0000	-	-	no	WD
BAL	DR	DO vs WD	0.0000	-	-	no	WD
BAL	EMR	DN vs DO	0.0000	-	-	no	DO
BAL	EMR	DN vs WD	0.0000	-	-	no	WD
BAL	EMR	DO vs WD	0.0000	-	-	no	WD
BAL	GLG	DO vs WD	-0.1376	0.2371	0.7113	no	DO
BAL	GLG	DN vs DO	0.0752	0.5192	1.0000	no	DO
BAL	GLG	DN vs WD	-0.0760	0.5147	1.0000	no	DN
BAL	GMG	DN vs DO	0.0000	-	-	no	DO
BAL	GMG	DN vs WD	0.0000	-	-	no	WD
BAL	GMG	DO vs WD	0.0000	-	-	no	WD
BAL	HARM	DO vs WD	0.7352	0.0000	0.0000	yes	WD
BAL	HARM	DN vs DO	-0.5872	0.0000	0.0000	yes	DN
BAL	HARM	DN vs WD	0.1432	0.2185	0.2185	no	WD
BAL	P95	DO vs WD	0.7304	0.0000	0.0000	yes	WD
BAL	P95	DN vs DO	-0.5448	0.0000	0.0000	yes	DN
BAL	P95	DN vs WD	0.1304	0.2626	0.2626	no	WD
MNS	DR	DN vs DO	0.0000	-	-	no	DO
MNS	DR	DN vs WD	0.0000	-	-	no	WD
MNS	DR	DO vs WD	0.0000	-	-	no	WD
MNS	EMR	DN vs DO	0.0000	-	-	no	DO
MNS	EMR	DN vs WD	0.0000	-	-	no	WD
MNS	EMR	DO vs WD	0.0000	-	-	no	WD
MNS	GLG	DN vs DO	-0.0056	0.9643	1.0000	no	DN
MNS	GLG	DN vs WD	-0.0584	0.6172	1.0000	no	DN
MNS	GLG	DO vs WD	-0.0560	0.6319	1.0000	no	DO
MNS	GMG	DN vs DO	0.0000	-	-	no	DO
MNS	GMG	DN vs WD	0.0000	-	-	no	WD
MNS	GMG	DO vs WD	0.0000	-	-	no	WD
MNS	HARM	DN vs WD	-0.7064	0.0000	0.0000	yes	DN
MNS	HARM	DO vs WD	-0.6056	0.0000	0.0000	yes	DO
MNS	HARM	DN vs DO	0.0568	0.6270	0.6270	no	DO
MNS	P95	DO vs WD	-0.5792	0.0000	0.0000	yes	DO
MNS	P95	DN vs WD	-0.5264	0.0000	0.0000	yes	DN
MNS	P95	DN vs DO	0.0296	0.8013	0.8013	no	DO

Wkld	Met	Pair	d	p MWU	p Holm	Sig	Best
RCS	DR	DN vs DO	0.0000	-	-	no	DO
RCS	DR	DN vs WD	0.0000	-	-	no	WD
RCS	DR	DO vs WD	0.0000	-	-	no	WD
RCS	EMR	DN vs DO	0.0000	-	-	no	DO
RCS	EMR	DN vs WD	0.0000	-	-	no	WD
RCS	EMR	DO vs WD	0.0000	-	-	no	WD
RCS	GLG	DN vs WD	0.2120	0.0682	0.2047	no	WD
RCS	GLG	DN vs DO	0.1168	0.3158	0.4117	no	DO
RCS	GLG	DO vs WD	0.1472	0.2059	0.4117	no	WD
RCS	GMG	DN vs DO	0.0000	-	-	no	DO
RCS	GMG	DN vs WD	0.0000	-	-	no	WD
RCS	GMG	DO vs WD	0.0000	-	-	no	WD
RCS	HARM	DO vs WD	0.2376	0.0410	0.1229	no	WD
RCS	HARM	DN vs DO	-0.1384	0.2344	0.4687	no	DN
RCS	HARM	DN vs WD	0.0880	0.4503	0.4687	no	WD
RCS	P95	DO vs WD	0.1760	0.1302	0.3907	no	WD
RCS	P95	DN vs DO	-0.0400	0.7329	0.7514	no	DN
RCS	P95	DN vs WD	0.1032	0.3757	0.7514	no	WD
SKB	DR	DN vs DO	0.0000	-	-	no	DO
SKB	DR	DN vs WD	0.0000	-	-	no	WD
SKB	DR	DO vs WD	0.0000	-	-	no	WD
SKB	EMR	DN vs DO	0.0000	-	-	no	DO
SKB	EMR	DN vs WD	0.0000	-	-	no	WD
SKB	EMR	DO vs WD	0.0000	-	-	no	WD
SKB	GLG	DN vs DO	-0.0296	0.8013	0.8605	no	DN
SKB	GLG	DN vs WD	-0.1240	0.2868	0.8605	no	DN
SKB	GLG	DO vs WD	-0.0960	0.4100	0.8605	no	DO
SKB	GMG	DN vs DO	0.0000	-	-	no	DO
SKB	GMG	DN vs WD	0.0000	-	-	no	WD
SKB	GMG	DO vs WD	0.0000	-	-	no	WD
SKB	HARM	DN vs DO	-0.3016	0.0094	0.0283	yes	DN
SKB	HARM	DN vs WD	-0.2256	0.0523	0.1046	no	DN
SKB	HARM	DO vs WD	0.0776	0.5059	0.5059	no	WD
SKB	P95	DN vs DO	-0.3520	0.0024	0.0073	yes	DN
SKB	P95	DN vs WD	-0.2064	0.0759	0.1517	no	DN
SKB	P95	DO vs WD	0.1712	0.1411	0.1517	no	WD

Table 4 reports all 72 pairwise rows using fixed abbreviations (workload/metric/policy codebook in caption), and the machine-readable export is `paper_submission_final_ros2/tables/table_pairwise_significance.csv`. Among the 72 tests, 10 satisfy Holm-adjusted $p < 0.05$. All significant rows are confined to `harm_weighted_cost` (5 rows) and `p95_latency_ms` (5 rows); no significant findings appear in `drop_rate`, `event_miss_rate`, `group_miss_gap`, or `group_latency_gap_ms`.

By workload:

- balanced: `drop_new` vs `drop_oldest` has lower `harm_weighted_cost_mean` (33.7644 vs 33.7675; Holm $p = 8.5 \times 10^{-7}$; $d = -0.5872$, large) and lower `p95_latency_ms_mean` (3.4508 vs 3.4743; Holm $p = 5.43 \times 10^{-6}$; $d = -0.5448$, large). `weighted_drop` vs `drop_oldest` is also lower on harm (33.7637 vs 33.7675; Holm $p < 10^{-8}$; $d = 0.7352$, large) and p95 latency (3.4448 vs 3.4743; Holm $p < 10^{-8}$; $d = 0.7304$, large).

- minority_spike: `weighted_drop` has higher `harm_weighted_cost_mean` than `drop_new` (41.3087 vs 41.3019; Holm $p < 10^{-8}$; $d = -0.7064$, large) and `drop_oldest` (41.3087 vs 41.3017; Holm $p = 3.7 \times 10^{-7}$; $d = -0.6056$, large). It also has higher `p95_latency_ms_mean` than `drop_new` (14.2404 vs 14.1666; Holm $p = 1.165 \times 10^{-5}$; $d = -0.5264$, large) and `drop_oldest` (14.2404 vs 14.1619; Holm $p = 1.83 \times 10^{-6}$; $d = -0.5792$, large). The `drop_new` vs `drop_oldest` comparisons are non-significant on both metrics.

- recovery_stress: no pairwise test reaches Holm significance.

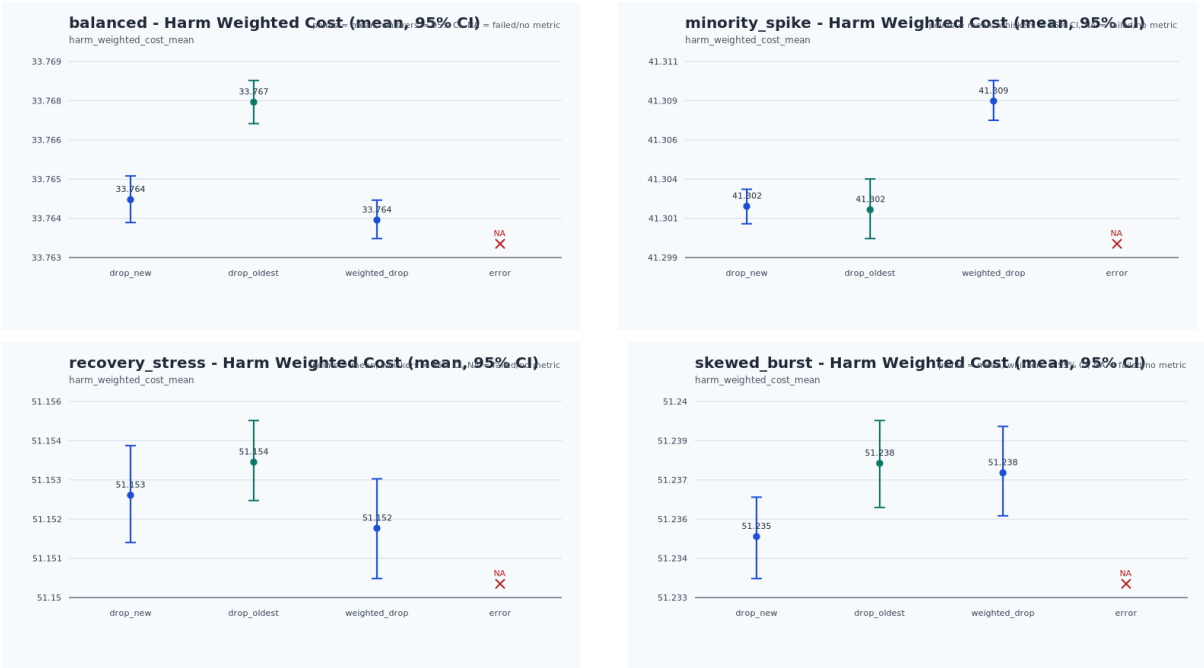

Figure 1: Per-workload `harm_weighted_cost_mean` point-and-whisker plots (mean with bootstrap 95% CI). `NA` marks failed/no-metric policies such as `error`.

- skewed_burst: `drop_new` vs `drop_oldest` has lower `harm_weighted_cost_mean` (51.2351 vs 51.2379; Holm $p = 0.0283$; $d = -0.3016$, small) and lower `p95_latency_ms_mean` (14.6361 vs 14.6828; Holm $p = 0.0073$; $d = -0.3520$, medium). Comparisons against `weighted_drop` are non-significant.

5 Discussion

This section separates observed facts from interpretation scope.

Observed in native runs. Policy choice changes rank order by workload (Table 3). Significant differences are concentrated in two metrics (`harm_weighted_cost`, `p95_latency_ms`) and three workloads (balanced, minority_spike, skewed_burst). Recovery_stress shows no Holm-significant pairwise differences; the largest observed mean differences are Δ`harm_weighted_cost` $= 0.00197$ and Δ`p95_latency_ms` $= 0.01953$.

Observed non-significance that is retained. `drop_rate` and `event_miss_rate` are identical across non-error policies within each workload. `group_miss_gap` remains 0.0 for all non-error cells. `group_latency_gap_ms` differs numerically but remains non-significant after Holm correction. These null findings are part of the result, not exclusions.

Interpretation boundary. The paper reports comparative behavior under this synthetic workload family and fixed metric definitions. No causal mechanism is claimed beyond observed associations. No claim is made that one policy is globally superior across all workloads.

Limitations. Workloads are synthetic, fairness weighting is configuration-dependent, and policy behavior is evaluated in single-host experimental setups. Therefore, external validity to different workload distributions or deployment constraints is not asserted.

6 Conclusion

Under a frozen $4 \times 4 \times 50$ protocol, native policy ranking is workload-dependent rather than uniform. Significant differences are limited to `harm_weighted_cost` and `p95_latency_ms`; other

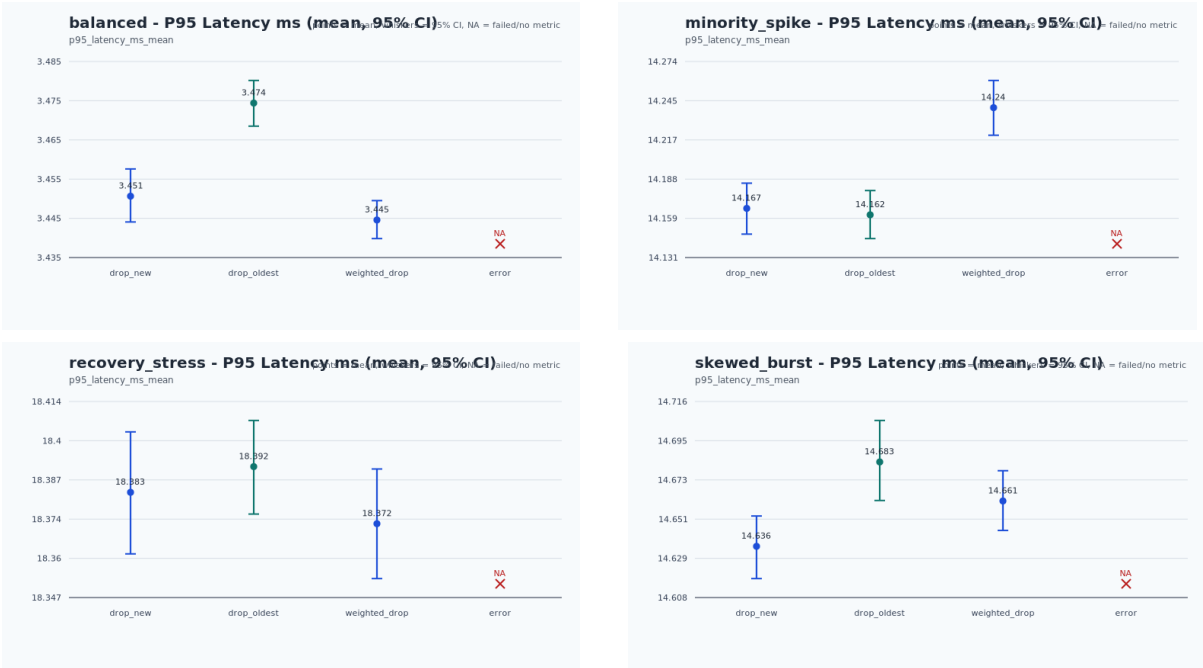

Figure 2: Per-workload `p95_latency_ms_mean` point-and-whisker plots (mean with bootstrap 95% CI). `event_miss_rate` and `drop_rate` are omitted here because they are policy-invariant within each workload.

tested metrics remain non-significant in this matrix. The `error` policy results are fully reported as failed-run outcomes.

The submission artifact set is reproducible and contract-checked, with ROS2 comparison kept as appendix-only evidence. Claims are restricted to measured comparisons in the fixed dataset and do not assert broader causal or universal behavior.

A Appendix: ROS2 Comparative Evidence

This appendix compares native and ROS2 transport baseline runs under the same schema and matrix shape ($4 \times 4 \times 50 = 800$ runs each). Core claims remain native-centric; this section is supportive evidence only.

From `ros2_comparison.csv`, all 12 comparable non-error workload-policy cells show lower native means for both `harm_weighted_cost` and `p95_latency_ms` (Table 5). The `error` policy is missing in both systems for comparison rows and is kept as explicit missing data.

From `ros2_reproducibility.csv`, both systems have equal failure rate (0.25) because `error` runs fail by design in both matrices. Reproducibility dispersion is higher in ROS2 baseline under this setup (`repro_cv_mean`=0.6326 vs 0.2901; `repro_cv_max`=1.6074 vs 0.5846). This appendix does not generalize to all ROS2 deployments; it only describes this transport baseline configuration.

B Reproducibility Package

Single source of truth for this manuscript: `outputs/experiments/backpressure_fairness/paper_submissio`

The package includes:

- frozen configs: `final_matrix_v1.yaml`, `analysis_spec_v1.yaml`,
- aggregate outputs: `aggregate/backpressure_aggregate.json` and CSV companions,

Table 5: Native vs ROS2 appendix comparison.

Workload	Policy	Native Harm	ROS2 Harm	Delta Harm	Winner	Native P95	ROS
balanced	drop_new	33.7644	100.9389	-67.1745	native	3.4508	64.13
balanced	drop_oldest	33.7675	101.1942	-67.4267	native	3.4743	68.48
balanced	error	-	-	-		-	-
balanced	weighted_drop	33.7637	101.6193	-67.8556	native	3.4448	72.22
minority_spike	drop_new	41.3019	106.9239	-65.6220	native	14.1666	94.85
minority_spike	drop_oldest	41.3017	106.8374	-65.5357	native	14.1619	92.29
minority_spike	error	-	-	-		-	-
minority_spike	weighted_drop	41.3087	106.6339	-65.3252	native	14.2404	84.17
recovery_stress	drop_new	51.1527	105.1598	-54.0070	native	18.3828	88.33
recovery_stress	drop_oldest	51.1537	104.9408	-53.7871	native	18.3916	80.60
recovery_stress	error	-	-	-		-	-
recovery_stress	weighted_drop	51.1518	103.8623	-52.7105	native	18.3720	68.02
skewed_burst	drop_new	51.2351	104.6816	-53.4465	native	14.6361	62.52
skewed_burst	drop_oldest	51.2379	105.7728	-54.5349	native	14.6828	77.71
skewed_burst	error	-	-	-		-	-
skewed_burst	weighted_drop	51.2375	105.1007	-53.8632	native	14.6612	70.29

- ROS2 comparison outputs: `ros2_comparison.json/csv`, `ros2_reproducibility.csv`,
- manuscript tables: `package/latex_tables/*.tex`,
- execution and QA records: `package/artifacts_manifest.json`, `package/repro_commands.md`, `package/submission_checklist.md`.

Interpreter split is required:

- native stages: conda `schnitzel-stream` Python with `PYTHONPATH=src`,
- ROS2 stages: `/usr/bin/python3` after sourcing `/opt/ros/humble/setup.bash`.

The final manifest for this artifact set reports: `ros2_baseline_stage_status=ok`, `ros2_compare_stage_st` and all configured quality gates as passed.